

Backup Types

- **Full Backup:** A full backup involves copying all data from a computer system to another location, such as an external hard drive, a cloud storage service, or a tape backup. This type of backup creates an exact copy of all files and folders, including system files, applications, and settings. A full backup is ideal for situations where the data is critical and cannot be lost.
- **Incremental Backup:** Incremental backups capture only the changes made to files and folders since the last backup was taken. These changes are stored in a separate file, called an incremental backup file, which is much smaller than a full backup file. Incremental backups require less storage space and are faster to perform than full backups. However, they can be slower to restore, as each incremental backup file must be applied to the previous backup to restore the data.
- **Differential Backup:** Differential backups capture all changes made to files and folders since the last full backup was taken. Unlike incremental backups, differential backups do not require multiple backup files to be restored. Instead, only the full backup and the latest differential backup file are required to restore the data. Differential backups are faster to restore than incremental backups, but they require more storage space and take longer to perform than incremental backups.
- **Copy Backup:** A copy backup is a type of full backup that creates an exact copy of all data on a computer system. Unlike a normal full backup, a copy backup does not reset the archive bit on each file. This means that a copy backup does not affect other backup types, such as incremental or differential backups. Copy backups are useful for creating a secondary copy of critical data that needs to be kept offsite.
- **Mirror Backup:** A mirror backup is a type of full backup that creates an exact copy of data on a computer system, including all files and folders, applications, and settings. Unlike a copy backup, a mirror backup resets the archive bit on each file, so it can be used in conjunction with other backup types, such as incremental or differential backups. Mirror backups are useful for creating a secondary copy of critical data that needs to be kept offsite.